

RESEARCH ARTICLE

Winners and losers under hydroxide-based ocean alkalinity enhancement in a Tasmanian plankton community

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Abstract

Ocean alkalinity enhancement (OAE) is an emerging carbon dioxide CO₂ removal approach for climate change mitigation and can be implemented with various alkaline materials that convert dissolved CO₂ into (bi)carbonates, enabling additional atmospheric CO₂ removal. A key knowledge gap is how alkaline materials affect marine life. This study investigated effects of OAE via sodium hydroxide (NaOH) on a coastal Tasmanian plankton community. Natural communities were enclosed within microcosms assigned to three groups: a control, an unequilibrated treatment (NaOH addition), and an equilibrated treatment (NaOH and sodium bicarbonate (NaHCO₃) addition). The unequilibrated treatment simulates carbonate chemistry changes before atmospheric CO₂ uptake and the equilibrated treatment the changes thereafter. Treatments increased alkalinity by ~25% (+500 μmol · kg⁻¹), theoretically enabling a 21% increase in the marine inorganic carbon sink. Hydroxide-based OAE had minimal effects on the plankton community in the equilibrated treatment, in which CO₂ and pH excursions were small. In the unequilibrated treatment, we observed a slight delay in the phytoplankton bloom, arguably because NaOH addition caused reorganization in the diatom community before the bloom reached its maximum chlorophyll *a* level. Although the community remained diatom-dominant, community composition was moderately different from the control and equilibrated treatments. The zooplankton community displayed no detectable change except for the invasive *Noctiluca scintillans*, which became less abundant in the unequilibrated treatment, arguably due to phytoplankton community shifts. We concluded changes in plankton community composition observed were relatively small compared to the rather extreme hydroxide-based alkalinity perturbation and the profound climatic benefit of such a CO₂ sink enhancement.

KEYWORDS

ocean alkalinity enhancement, phytoplankton

Abbreviations: CDR, carbon dioxide removal; Chl *a*, chlorophyll *a*; DIC, Dissolved inorganic carbon; DSi, dissolved silicate; FSC-A, forward scatter area; GAM, generalized additive model; NaHCO₃, sodium bicarbonate; NaOH, sodium hydroxide; OA, ocean acidification; OAE, ocean alkalinity enhancement; pCO₂, partial pressure of carbon dioxide; PE, polyethylene; pH_T, total scale pH; SEM, scanning electron microscopy; TA, total alkalinity.

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INTRODUCTION

Many nations have committed to keeping global warming below 1.5–2°C (Intergovernmental Panel on Climate Change [IPCC], 2018; United Nations, 2016). Achieving this goal requires both a rapid decrease in CO₂ emissions and the active removal of CO₂ from the atmosphere (IPCC, 2018). Atmospheric CO₂ removal (CDR) will likely require a diverse portfolio of terrestrial and marine CDR methods to collectively remove on the order of 10 gigatons of CO₂ per year by 2050 (IPCC, 2023).

Ocean alkalinity enhancement (OAE) is an emerging CDR method of the marine portfolio that is inspired by the natural weathering of alkaline minerals in aqueous environments (Hartmann et al., 2013). Natural rock weathering sequesters CO₂ through the slow breakdown and dissolution of alkaline minerals over geological timescales (i.e., >10,000 years) in aqueous media. During the dissolution process, alkaline rocks neutralize H⁺ ions and shift the concentration of inorganic carbon species in seawater from CO₂ toward bicarbonate (HCO₃[−]) and carbonate (CO₃^{2−}), measurable as an increase in total alkalinity (TA). The shift toward HCO₃[−] and CO₃^{2−} enables additional storage of atmospheric CO₂ in seawater with a residence time of ~100,000 years (Middelburg et al., 2020). However, due to the inertia of weathering processes, natural CO₂ sequestration through these processes in the Earth's system cannot keep pace with the rate of anthropogenic CO₂ emissions. As such, several OAE methods have been devised to anthropogenically enhance the rate of H⁺ neutralization in marine systems. One widely considered OAE method uses electric energy to electrochemically split seawater into a strong acid (mostly hydrochloric acid) and a strong base (mostly sodium hydroxide, NaOH; Eisaman et al., 2023). The acidic stream needs to be neutralized or stored safely outside seawater while the alkaline stream containing NaOH is released back into the ocean. The addition of NaOH to seawater leads to a rapid increase in alkalinity, which converts seawater CO₂ into HCO₃[−] and CO₃^{2−} within minutes. This conversion is substantially faster than atmospheric CO₂ influx through the air–sea interface, which takes weeks to years (Jones et al., 2014; Zhou et al., 2024). As such, enhancing alkalinity via NaOH addition leads to different carbonate chemistry conditions before and after the equilibration of seawater with atmospheric CO₂ (commonly referred to as “unequilibrated” and “equilibrated” OAE; Bach et al., 2019). The unequilibrated state is characterized by more extreme changes in carbonate chemistry conditions (low CO₂ and elevated pH), while carbonate chemistry in the equilibrated state is similar to the unperturbed (control) state that existed before alkalinity was added (Ferderer et al., 2022).

Although OAE is a widely considered method for achieving atmospheric CDR at a gigaton scale, its potential impact on marine ecosystems is currently not well constrained. The alteration to water chemistry presented by OAE has the potential to affect various organisms, including phytoplankton and zooplankton. This study aimed to investigate how OAE influences the composition of a coastal plankton community. The experiment consisted of a sampled plankton community enclosed in nine microcosms and split into three groups: a control group, an equilibrated treatment (addition of NaOH and sodium bicarbonate, NaHCO₃), and an unequilibrated treatment (addition of NaOH). The microcosms were kept in a temperature-controlled room, mimicking natural temperature and light conditions over the 21-day experiment. The experiment allowed us to investigate changes in plankton communities that occurred between treatments. The experiment largely followed the protocol from the Ocean Alkalinity Enhancement Pelagic Impact Intercomparison Project (OAPEIIP), which aims to identify common OAE-responses across a range of environments globally (Bach et al., 2024).

MATERIALS AND METHODS

Microcosm setup and experimental design

Water samples were collected from the Derwent Estuary in Hobart, Tasmania on March 28, 2023. The collection took place next to the Institute of Marine and Antarctic Studies (42.88584° S, 147.33479° E), as shown in Figure 1a. The biogeochemistry of the Derwent Estuary is influenced by nutrient inputs from terrestrial, industrial, and agricultural sources, with moderately stable pH levels that have ranged from 7.8 to 8.36 over a 5-year period (2020–2025; Figure 1a). The nine microcosms used in this experiment consisted of 60-L cylindrical plastic containers equipped with an upper threaded lid and a bottom valve (Ferderer et al., 2022; Figure 1c). The microcosms were fitted with 2-mm mesh over both openings before filling to exclude large zooplankton grazers. The open microcosms were attached to a crane and slowly lowered ~1 m into the water, 2 m away from the wharf wall, to be filled (Bach et al., 2024). Once full, the bottom valve was manually closed using a rope mechanism, and the microcosms were raised to the surface. Filling occurred in a randomized order (M1, M2, M4, M9, M5, M6, M8, M3, and M7) over a 45-min period. The microcosms were left with ~20 cm of headspace, sealed, dried, and weighed with seawater (ranging from 52.22 kg to 55.59 kg). The microcosms were then transported into the laboratory for experimental setup. Microcosms are equipped with small closeable openings on the lid to allow for sampling, minimizing exchange with the external environment. A SonTek Castaway CTD was also deployed

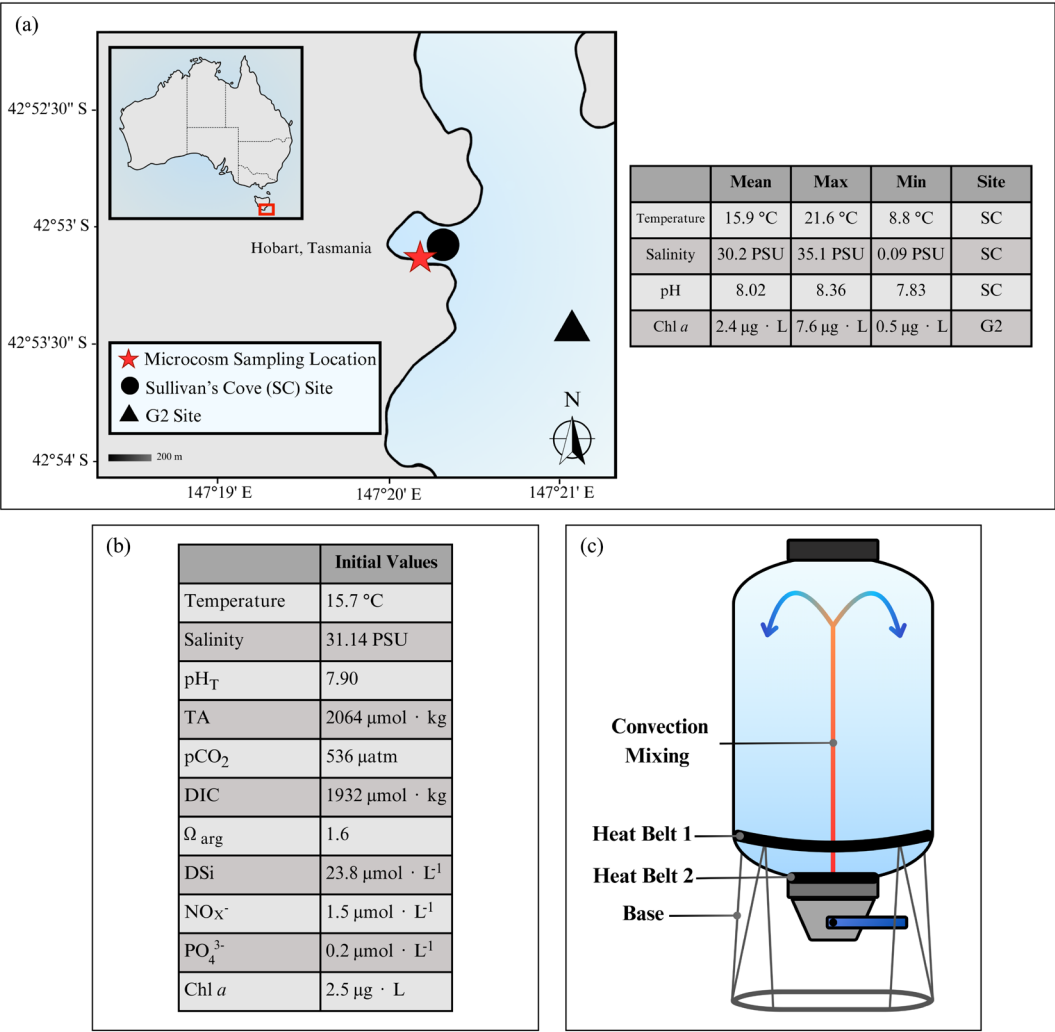


FIGURE 1 (a) Map detailing the sampling location of the experiment (42.88584° S, 147.33479° E). The map also details sampling locations of for biogeochemical parameter measurements that took place in the Derwent Estuary over a 5-year span (2020–2025). The mean, max, and min values of temperature, salinity, pH, and chlorophyll *a* are listed; data provided courtesy of the Derwent Estuary Program. CTD measurements of temperature, salinity, and pH were taken from the Sullivan's Cove site (42.88515° S, 147.3369° E), and chlorophyll *a* measurements took place in lab with bottle samples from the G2 site (42.89098° S, 147.3505° E; Derwent Estuary Program, 2025). (b) Table detailing the initial parameter measurements taken on day 0, prior to any alkalinity manipulation. Initial parameters include temperature, salinity, pH_T (pH total scale), TA (total alkalinity), pCO₂ (partial pressure of carbon dioxide), DIC (dissolved inorganic carbon), Ω_{arg} (aragonite saturation state), DSi (dissolved silicate), NO_x⁻ (nitrate + nitrite), PO₄³⁻ (phosphate), and Chl *a* (chlorophyll *a*). (c.) Diagram detailing microcosm setup, including the location of heat belts and the process of convection mixing provided by the heat belts.

during the time of collection to monitor temperature and salinity conditions.

Immediately after the filling procedure, all nine microcosms were placed into a temperature-controlled room set to 8°C. Each microcosm was equipped with two 30-watt heat belts, one around the valve base and one around the upper stand of the base (Figure 1c). The room's temperature allowed the microcosms to reach an anticipated incubation temperature of 14°C once the heat belts were activated. The belts were used to create convection currents and simulate natural water movement as the water was warmed at the bottom with the heat belts and cooled from the outside due to the cooler room temperature (Ferderer et al., 2022; Figure 1c). Ten,

5-m LED light strips (SMD3014 waterproof LEDs with 204 LEDs per meter in cool white color of 6000–6500 K) were positioned in a U-shape along the walls of the room, providing ~200 µmol photons · m⁻² · s⁻¹ inside the microcosms. The lights were set on a timer, following a 12:12 h light:dark cycle. To mitigate the effects of inconsistencies in light and temperature, all microcosms were systematically shifted around the room daily.

The nine microcosms were divided into three groups: the control (M1, M4, and M7), the unequilibrated treatment (M2, M5, and M8), and the equilibrated treatment (M3, M6, and M9). Initial parameter measurements were taken prior to any alkalinity addition (Figure 1b). The control group received no alkalinity enhancement.

The equilibrated treatment received $\sim 80 \mu\text{mol}$ of sodium hydroxide (NaOH - 1 mol liquid from Merck, tritipur—reagent grade) per kg of seawater, along with a $\sim 420 \mu\text{mol}$ addition of sodium bicarbonate (NaHCO_3 —Sigma-Aldrich, reagent grade) per kg of seawater, and the unequilibrated treatment received $500 \mu\text{mol}$ of sodium hydroxide (NaOH) per kg of seawater. Microcosms were stirred immediately after alkalinity additions and sampling began after a 10-min waiting period at 3:00 p.m. (local time). In the following sampling days, microcosms were sampled between 7:30 a.m. and 9:30 a.m. for 21 days. The sampling frequency varied between parameters (see next section).

Sampling and analytical procedures

Temperature and pH were measured in situ daily using the same pH electrode (Metrohm Aquatrode Plus). All other measurements were obtained from samples collected directly from microcosms either through pipetting (flow cytometry) or with a peristaltic pump. Chlorophyll *a* (Chl *a*), light microscopy, and scanning electron microscopy (SEM) samples were pumped through silicon tubes and into clean bottles prerinsed with sample. Nitrate/nitrite (NO_x^-), phosphate (PO_4^{3-}), dissolved silicate (DSi), and TA samples were pumped through a Tygon tube attached to a $0.2\text{-}\mu\text{m}$, 25-mm nylon syringe filter and into clean, prerinsed polyethylene (PE) bottles.

Dissolved inorganic nutrients, DSi, NO_x^- , and PO_4^{3-} samples were collected every other day and analyzed within 3–7 h of sampling using photometric methods described by Hansen and Koroleff (1999).

Samples for Chl *a* and SEM were filtered within 2 h of sampling using a vacuum filtration system at a vacuum pressure of less than 200 mbar. Filter funnels were rinsed with double deionized water in between samples.

Chlorophyll *a* samples (150 mL) were collected every other day and filtered onto 25-mm glass fiber filters (GF/F). These filters were placed into test tubes, which were wrapped in foil and stored at -80°C for no longer than 3 months. For fluorometric analyses, 10 mL of 100% methanol was added to the test tubes containing the filters. The tubes were shaken and placed into a 4°C refrigerator for 12–16 h. Samples were subsequently centrifuged for 2 min at 2000 rpm and 5°C . The supernatant was decanted into a clean glass tube before being measured on a Turner 10-AU fluorometer. An addition of 0.15 mL of 0.1 M HCL was then added to the fluorometer tube before it was inverted and measured again, following Evans et al. (1987).

Scanning electron microscopy samples (30 mL) were collected four times and filtered onto $0.2\text{-}\mu\text{m}$, 25-mm polycarbonate filters, which were stored in plastic petri dishes. Filters were then dried at 60°C for 1 h with the

lids open, then wrapped in aluminum foil and stored at room temperature. Filters were glued onto SEM stubs and sputtered with a layer of 5 nm platinum. These filters were photographed and analyzed under SEM for taxonomic identification of species, which were originally quantified with the light microscope (Table S1).

Phytoplankton light microscopy samples were collected every 4 days, as well as day 2, in 150-mL glass bottles and fixed with 1.5 mL acidic Lugol's solution. A total of 5 mL of sample was then added to a Utermöhl settling chamber and allowed to settle for 16 h. Samples were viewed under 4x and 20x magnification. The entirety of the chamber was observed under 4x magnification to record all large and rare phytoplankton individuals. At 20x magnification, phytoplankton specimens larger than $10 \mu\text{m}$ were recorded, up to 300 individuals. Phytoplankton light microscopy methods were adapted from procedures outlined by Karlson et al. (2010). Identified phytoplankton groups were defined by highly abundant or ecologically common genera and/or species. Phytoplankton that did not fit these parameters were grouped as “Others.”

Zooplankton were sampled three times during the experiment using a miniature plankton net (Guo et al., 2024). Each microcosm was gently stirred with a plastic paddle before the net was lowered to the bottom. The net was then raised at $\sim 30 \text{ cm} \cdot \text{s}^{-1}$ over a vertical distance of $\sim 60 \text{ cm}$. The nets were rinsed into plastic jars and treated with an addition of 10% formalin in seawater. Samples were stored in the dark at room temperature until analysis. Before microscopic viewing, samples were rinsed of formalin using a seine. The sample was placed into a Bogarov counting chamber, where the entire sample was observed, and all zooplankton groups counted and identified.

Flow cytometry samples were collected every other day by pipetting directly from the top ($\sim 5 \text{ cm}$ depth) of each microcosm and placing the samples into 2-mL cryovials. For phytoplankton samples, 1.75 mL of sample from each microcosm was added to separate cryovials, along with $50 \mu\text{L}$ of formaldehyde/hexamine solution. For bacteria samples, 1 mL of microcosm sample was added to separate cryovials, along with $20 \mu\text{L}$ of glutaraldehyde. All samples were stored at 4°C for 20 min before being flash frozen in liquid nitrogen and stored at -80°C . After 20 weeks, samples were thawed and analyzed with a Cytex Aurora flow cytometer. Bacteria samples were stained with SYBR Green I prior to analysis to identify them via their DNA's green fluorescence and the absence of autofluorescence. Phytoplankton groups were distinguished based on autofluorescence and scatter properties (Ferderer et al., 2022; Guo et al., 2024). The results were analyzed using CYTEK SpectroFlo v2 software to gate different factions within the communities and calculate the contribution of different groups to obtain the community composition (Figures S1 and S2).

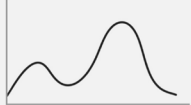
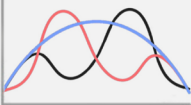
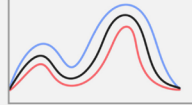
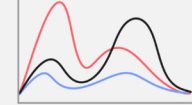
Model 1	Model 2	Model 3	Model 4
Maintains uniform temporal trends across all treatments	Variation in temporal trends and consistent absolute values	Variation in absolute values and consistent temporal trends	Variation in temporal trends and absolute values
			
Nutrients (AIC)			
Chl <i>a</i> (90.42)	NO _x (148.3) _(E, U)	DSi (88.51) _(C, U) PO ₄ ³⁻ (-70.36) _(C, U)	
Phytoplankton Light Microscopy Abundance (AIC)			
Flagellates (501.7) Others – Phyto (234.6) <i>Mesodinium rubrum</i> (445.2) <i>Dactyliosolen</i> spp. (257.1) <i>Cylindrotheca</i> spp. (468.7)	<i>Ceratulina pelagica</i> (297.7) _(C, E, U)		<i>Pseudo-nitzschia</i> spp. (616.9) _(C, U) <i>Skeletonema</i> spp. (296.1) _(C, U) <i>Leptocylindrus</i> spp. (192.5) _(C, U) <i>Chaetoceros</i> spp. (512.6) _(C, U) <i>Thalassionema</i> spp. (489.1) _(C, E, U)
Flow Cytometry Abundance (AIC)			
Cryptophytes (159.1) Nanophytoplankton 1 (235.7) <i>Synechococcus</i> (146.3) Bacteria (210.6)	Picoeukaryotes (186.9) _(C, E, U)	Nanophytoplankton 2 (93.99) _(C, E, U)	

FIGURE 2 Generalized additive model (GAM) results. The Akaike information criteria of the preferred model is displayed below the description of the model used. The model with the lowest AIC was considered the preferred model unless the difference in AIC was <2, in which case Model 1 (the simplest model) was selected as the default. Significant results are labeled with C (control), E (equilibrated), and/or U (unequilibrated) to denote which treatments showed significant differences from each other (e.g., CE indicates a significant difference between the control and equilibrated treatment). *Rhizosolenia* spp. was not included in the table as it did not fit any of the assumptions.

Total alkalinity samples were collected every fourth day and stored at 6°C until analysis within 60 days. The TA analysis of 50–60 mL samples followed the open cell titration described by Dickson et al. (2007) using a Metrohm 862 Compact Titrator. Titrations were accuracy controlled using certified reference material (CRM, batch #200) provided by Prof. Dickson's laboratory. The collected titration curves were analyzed in Python using the “calkulate” function of the PyCO2SYS toolbox (Humphreys et al., 2022).

Potentiometric pH was calibrated to the total scale (pH_T) at the start of the experiment using certified reference material (TRIS buffer, batch T37) provided by Prof. Dickson's laboratory and following the methods outlined in Dickson et al. (2007). Using the Seacarb package in R (Gattuso et al., 2021), pH_T, temperature, salinity, phosphate, silicate, TA, and equilibrium constants K₁ and K₂ from Lueker et al. (2000) (as recommended by Orr et al., 2015) were used to determine all other relevant carbonate chemistry parameters (pCO₂, DIC, and Ω_{arg}). Data interpolation was conducted on TA, phosphate, and silicate in order to match the temporal resolution of these measurements with the daily measurements of temperature and pH_T. This interpolation allowed us to calculate daily carbonate chemistry conditions with Seacarb. All calculations used

the default setting for carbonate chemistry equilibrium constants recommended in Seacarb.

Data analysis

Phytoplankton and zooplankton biovolume from light microscopy were determined by measuring the size dimensions of genera/species and converting these into volume using shape-specific geometric formulas (Table S2). The average volume determined for each genus was multiplied by its abundance to calculate the community biovolume. Phytoplankton and bacteria biovolume from flow cytometry was calculated using an equation derived from the relationship between biovolume and forward scatter area (FSC-A) measurements for the phytoplankton (bacteria) groups (Selfe, 2022):

$$\begin{aligned}
 & \text{(Flow cytometry) biovolume} \\
 & = \text{cell number count} \times \left(\frac{\text{FSC-A}}{10248} \right)^{2.14}
 \end{aligned}$$

The analysis of Chl *a*, NO_x⁻, PO₄³⁻, phytoplankton abundance via light microscopy, as well as phytoplankton and bacterial abundance via flow cytometry, was

conducted using generalized additive models (GAMs; Figure 2). These models were fit using the *mgcv* package in RStudio (Wood, 2012). To determine the best-fit model, four separate models were applied. All models included “day” fit as a smooth term and “microcosm” nested within “day” fit as a random factor smooth to account for potential variation between microcosms within treatments. Model 1 described the scenario where there was no significant difference in the temporal trends of a given parameter over the extent of the experiment; Model 1: $Y = s(\text{Day}) + s(\text{Day}, \text{Microcosm})$. Model 2 described a scenario where there was variation in temporal trends but consistent absolute values; Model 2: $Y = s(\text{Day}, \text{by} = \text{Treatment}) + s(\text{Day}, \text{Microcosm})$. Model 3 described the scenario that allowed for significant differences in variation of absolute values across the treatments, but temporal trends were kept consistent; Model 3: $Y = s(\text{Day}) + s(\text{Day}, \text{Microcosm}) + \text{Treatment}$. Model 4 described a scenario that allowed for significant differences between treatment groups in both temporal trends and absolute values; Model 4: $Y = s(\text{Day}, \text{by} = \text{Treatment}) + s(\text{Day}, \text{Microcosm}) + \text{Treatment}$. Model assumptions were visually assessed, and appropriate data was log or square root transformed.

To investigate the effects of the treatments on zooplankton abundances, we employed a linear mixed-effects model with treatment and zooplankton genera included as a two-way interaction and abundance data (square root transformed) as the dependent variable. To account for potential variability between sampling days and between mesocosms within each day, random effects were specified for treatment at the level of day and microcosm. Pairwise comparisons of the estimated marginal means, using the package *emmeans* (Lenth, 2023), were used to assess significant differences across the levels of treatment and zooplankton groups, enabling the evaluation of the interaction. All statistical analyses were conducted in RStudio v 2023.6.1.524 (R Core Team, 2023).

RESULTS

The carbonate system

Total alkalinity was initially $2066 \pm 3.6 \mu\text{mol} \cdot \text{kg}^{-1}$ in the control, $2063 \pm 1.3 \mu\text{mol} \cdot \text{kg}^{-1}$ in equilibrated treatment, and $2062 \pm 4.4 \mu\text{mol} \cdot \text{kg}^{-1}$ in the unequilibrated treatment (Figure 3a; Table S3, uncertainties here and hereafter refer to standard deviation between the three replicates). Values were calculated using the average of the three microcosms for each treatment. Immediately after the addition of alkalinity sources (NaOH, NaHCO_3), we measured elevated TA in both the equilibrated ($2557 \pm 5.6 \mu\text{mol} \cdot \text{kg}^{-1}$) and unequilibrated ($2568 \pm 4.6 \mu\text{mol} \cdot \text{kg}^{-1}$) treatments.

Note that TA measurements for microcosm 9 (equilibrated treatment) failed on day 0.5 (i.e., the sample taken directly after NaOH/ NaHCO_3 addition) and microcosm 2 (unequilibrated treatment) on day 20. For those two datapoints, we used averages of the other two replicates from the same day for carbonate chemistry calculations. Concentrations of TA remained relatively stable in both treatments and in the unperturbed control throughout the experimental period.

The pH_T was initially 7.91 ± 0.02 in the control, 7.91 ± 0.01 in the equilibrated treatment, and 7.91 ± 0.04 in the unequilibrated treatment (Figure 3b; Table S3). Immediately after alkalinity additions, pH_T values increased in both the equilibrated (8.04 ± 0.004) and the unequilibrated (8.65 ± 0.01) treatments. From this point, pH_T increased to 8.1 ± 0.03 in the control on day 6, 8.16 ± 0.02 in the equilibrated treatment on day 6, and 8.69 ± 0.02 in the unequilibrated treatment on day 6. After day 8, pH_T gradually declined in the control and in both treatments.

The partial pressure of carbon dioxide (pCO_2) was initially $535.8 \pm 22 \mu\text{atm}$ in the control, $525.6 \pm 9 \mu\text{atm}$ in equilibrated treatment, and $548 \pm 58 \mu\text{atm}$ in unequilibrated treatment (Figure 3c; Table S3). Immediately after alkalinity additions, pCO_2 decreased in both the unequilibrated ($79.6 \pm 4 \mu\text{atm}$) and equilibrated ($432.9 \pm 50 \mu\text{atm}$) treatments. The control reached its lowest pCO_2 level of $317.6 \pm 36 \mu\text{atm}$ on day 7, followed by a gradual increase to $480.7 \pm 36 \mu\text{atm}$ by day 20. The equilibrated treatment reached its lowest pCO_2 level of $337.2 \pm 16 \mu\text{atm}$ on day 6, followed by a gradual increase to $458.9 \pm 11 \mu\text{atm}$ by day 20. The unequilibrated treatment reached its lowest pCO_2 measurement of $70.2 \pm 4 \mu\text{atm}$ on day 6, followed by a gradual increase to $110.6 \pm 8 \mu\text{atm}$ by day 20. Declining pCO_2 during the bloom and increasing pCO_2 after the bloom are reflected in the Chl *a* trend, reflecting photosynthetic buildup and respiratory decline of biomass.

Dissolved inorganic carbon (DIC) was initially $1934 \pm 4 \mu\text{mol} \cdot \text{kg}^{-1}$ in the control, $1928 \pm 1 \mu\text{mol} \cdot \text{kg}^{-1}$ in the equilibrated treatment, and $1933 \pm 10 \mu\text{mol} \cdot \text{kg}^{-1}$ in the unequilibrated treatment (Figure 3d; Table S3). Immediately after alkalinity additions, DIC was $1931 \pm 7 \mu\text{mol} \cdot \text{kg}^{-1}$ in the control, $2341 \pm 5 \mu\text{mol} \cdot \text{kg}^{-1}$ in the equilibrated treatment, and $1936 \pm 12 \mu\text{mol} \cdot \text{kg}^{-1}$ in the unequilibrated treatment. Dissolved inorganic carbon decreased in the control, reaching its lowest value of $1858 \pm 18 \mu\text{mol} \cdot \text{kg}^{-1}$ on day 7 remaining there for 3 days before gradually increasing up to $1921 \pm 13 \mu\text{mol} \cdot \text{kg}^{-1}$ by day 20. The equilibrated treatment increased up to 2341 ± 5 immediately after alkalinity addition, gradually decreased to $2287 \pm 8 \mu\text{mol} \cdot \text{kg}^{-1}$ on day 6, and then increased up to $2350 \pm 5 \mu\text{mol} \cdot \text{kg}^{-1}$ by day 20. The unequilibrated treatment gradually increased, ending at $1994 \pm 86 \mu\text{mol} \cdot \text{kg}^{-1}$ on day 20.

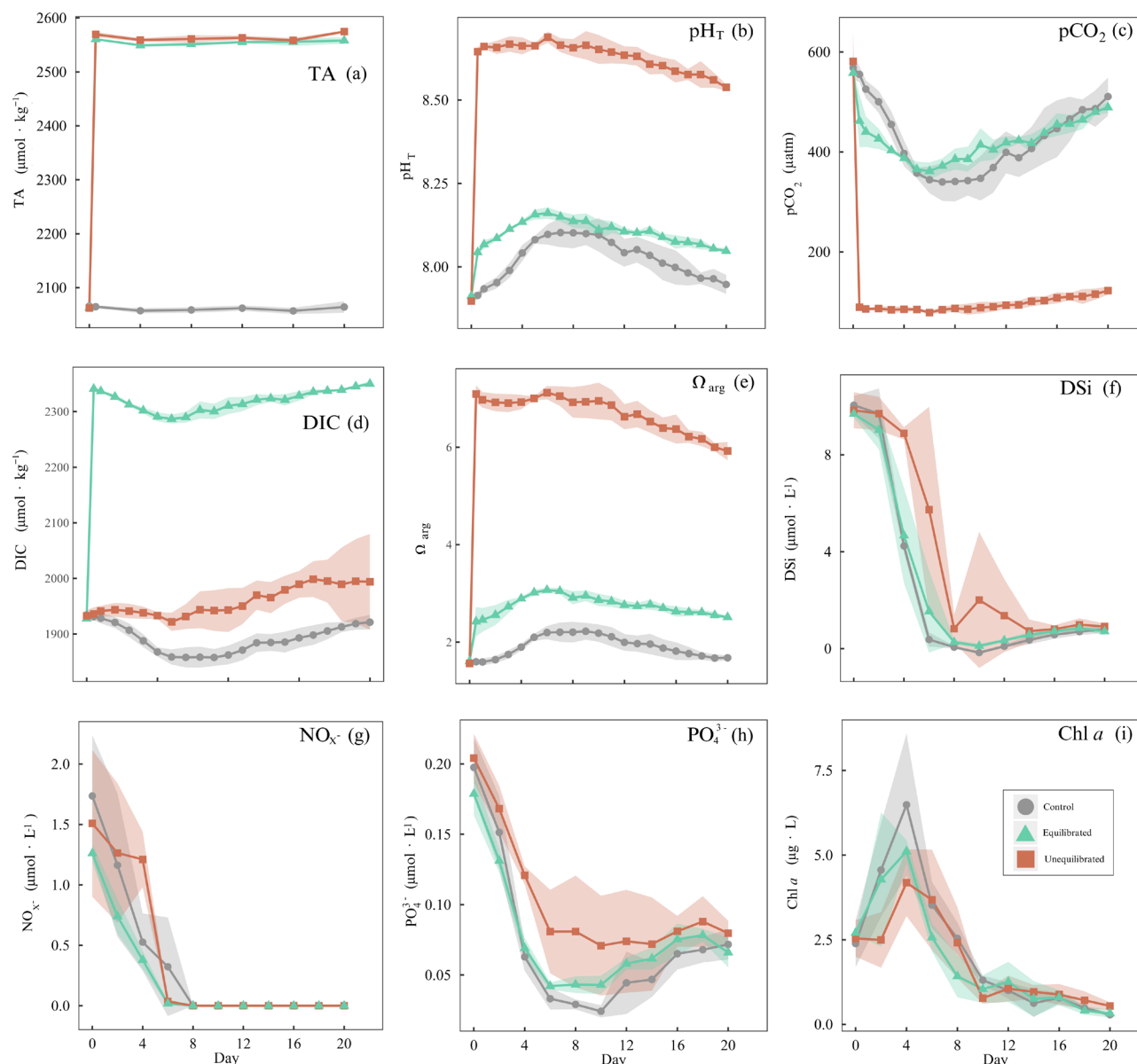


FIGURE 3 Temporal trends in carbonate chemistry and dissolved inorganic nutrient conditions. (a) TA (total alkalinity); (b) pH_T (pH total scale); (c) pCO_2 (partial pressure of carbon dioxide); (d) DIC (dissolved inorganic carbon); (e) Ω_{arg} (aragonite saturation state); (f) DSI (dissolved silicate); (g) NO_x^- (nitrate + nitrite); (h) PO_4^{3-} (phosphate); (i) Chl *a* (chlorophyll *a*). Standard deviation is represented by ribbon shading around the mean trendlines.

The saturation state for aragonite (Ω_{arg}) was initially 1.6 ± 0.1 in the control, 1.6 ± 0.02 in the equilibrated treatment, and 1.6 ± 0.1 in the unequilibrated treatment (Figure 3e; Table S3). The control increased up to a maximum of 2.2 ± 0.1 on day 6, remained steady for 4 days, and then declined to a value of 1.7 ± 0.1 by day 20. The equilibrated treatment reached 2.4 ± 0.3 directly after the alkalinity addition and remained mostly stable, ending with a slight increase to 2.5 ± 0.01 on day 20. Immediately after the alkalinity addition, the unequilibrated treatment increased to 7.1 ± 0.2 ; however, the Ω_{arg} of the unequilibrated treatment then gradually declined, ending at 5.9 ± 0.2 on day 20.

Dissolved inorganic nutrients

Dissolved inorganic nutrient concentrations (DSi; Nitrate + Nitrite, NO_x^- ; and Phosphate, PO_4^{3-}) began decreasing immediately in the control and equilibrated treatment (Figure 3f–h; Table S3). There were little to no significant differences in the drawdown of nutrients between the control and equilibrated treatment, as qualified by statistical analysis. In contrast, there was a significant difference in the drawdown of nutrients (DSi, NO_x^- , and PO_4^{3-}) in the unequilibrated treatment. There was an initial delay in the drawdown of NO_x^- in the unequilibrated treatment; however, by day 8, concentrations of NO_x^- had dropped

below detection limits in all microcosms (Figure 3f; Table S3). In contrast, the drawdown of DSI and PO_4^{3-} in the unequilibrated exhibited similar temporal patterns to other treatments; however, there was less nutrient drawdown, most evident within the first 8 days. Dissolved silicate approached limits of detection by day 8 in the control and equilibrated treatment but remained detectable until day 14 in the unequilibrated treatment (Figure 3e; Table S3). PO_4^{3-} levels decreased until day ~10, before slowly increasing over days 11–20 across all treatments (Figure 3g; Table S3).

Chlorophyll a

Initial Chl *a* concentrations were $2.4 \pm 0.7 \mu\text{g} \cdot \text{L}^{-1}$ in the control, $2.7 \pm 0.1 \mu\text{g} \cdot \text{L}^{-1}$ in the equilibrated treatment, and $2.5 \pm 0.5 \mu\text{g} \cdot \text{L}^{-1}$ in the unequilibrated treatment (Figure 3i; Table S3). All treatments peaked on day 4 with Chl *a* values of $6.5 \pm 2.1 \mu\text{g} \cdot \text{L}^{-1}$ in the control, $5.1 \pm 0.3 \mu\text{g} \cdot \text{L}^{-1}$ in the equilibrated treatment, and $4.2 \pm 1.0 \mu\text{g} \cdot \text{L}^{-1}$ in the unequilibrated treatment. Although Chl *a* peaked at the same day in all treatments, the increase in Chl *a* was delayed in the unequilibrated treatment after the onset of the experiment (days 0–2; Figure 3i). In all treatments, Chl *a* concentrations decreased after day 4 and approached limits of detection by day 20.

Plankton community composition

Phytoplankton

Communities within the microcosms revealed 12 main groups of phytoplankton detectable by microscopy. The majority of the groups were diatoms, with the exception of flagellates, “Non-Diatom (Flagellates)”; others, “Diatoms and Non-Diatoms” and *Mesodinium rubrum*, “Ciliate.” Most diatom species peaked in abundance during the Chl *a* peak (~day 4), with the exceptions of *Cerataulina pelagica*, *Cylindrotheca* spp., and *Dactylosolen* spp. that peaked around day 12 and *Thalassionema* spp. that peaked around day 8. Additionally, six distinct groups were identified via flow cytometry including *Synechococcus*, cryptophytes, picoeukaryotes, nanophytoplankton 1, nanophytoplankton 2, and bacteria.

The abundance of phytoplankton groups in the control versus equilibrated treatment showed significant mixed effects in *Thalassionema* spp. and picoeukaryotes (Figure 2). Within the equilibrated treatments, there was a decreased abundance in *Thalassionema* spp. (~days 4–12) and an increased abundance in picoeukaryotes (~days 12–15; Figures 4h and 5e). The biovolume of the equilibrated treatment showed the

largest increase in the diatom *Pseudo-nitzschia* spp., especially post-bloom (after day 9; Figure 6a,b).

The control versus unequilibrated treatment showed mixed effects in the groups *Pseudo-nitzschia* spp., *Skeletonema* spp., *Leptocylindrus* spp., *Chaetoceros* spp., *Cerataulina pelagica*, *Thalassionema* spp., picoeukaryotes, and nanophytoplankton 2 (Figure 2). Within the unequilibrated treatment, there was a decrease in the abundance of *Pseudo-nitzschia* spp. (~days 1–19), *Skeletonema* spp. (~days 1–15), *Leptocylindrus* spp. (~days 1–15), *Chaetoceros* spp. (~days 1–15), *Cerataulina pelagica* (~days 1–19), *Thalassionema* spp. (~days 1–19), picoeukaryotes (~days 1–5), and nanophytoplankton 2 (~days 3–13; Figures 4a,b,d–f,h and 5e,i). However, there was also an increase in the abundance of picoeukaryotes (~days 9–18; Figure 5e). When compared to the control, the biovolume of the unequilibrated treatment showed a decrease in some diatom groups and an increase in others. There was a decreased biovolume in the diatoms *Pseudo-nitzschia* spp., *Skeletonema* spp., and *Chaetoceros* spp. throughout the entire experiment (Figure 6a,c). There was also decreased biovolume in picoeukaryotes (~days 1–5) and nanophytoplankton 2 (~days 5–13) (Figure 5f,j). There was an increase in *Leptocylindrus* spp., as it made up a large amount of the biovolume until it dropped off around day 15, when the diatoms *Cerataulina pelagica*, *Cylindrotheca* spp., and *Dactylosolen* spp. increased in biovolume (Figure 6a,c). Additionally, there was an increase in the biovolume of cryptophytes (~days 2–15; Figure 5d).

Zooplankton

Four main groups of zooplankton were identified in the experiment. These included *Noctiluca scintillans* (heterotrophic), *Penilia* spp. (cladoceran), *Calanoida* spp. (copepod), and others. The linear mixed effects model revealed no difference in zooplankton community abundance across the three treatments, although there were strong differences in abundances between genera (linear mixed effects models – genera: $df=3,72$, f value=65.25, p value <0.001; Table S4, Table S5). Post hoc testing revealed *Noctiluca* abundances in the unequilibrated treatment to be significantly lower than the control (Post hoc test assessing the variation in zooplankton abundance within a genus as a function of the treatment: Control – Unequilibrated, group=*Noctiluca*, emtend=1.76, standard error=0.67, t ratio=2.61, p value=0.04), although no significant difference was observed between the equilibrated and control or equilibrated and unequilibrated (Table S6). No other genera displayed significant changes in abundances between the experimental treatments (Table S6).

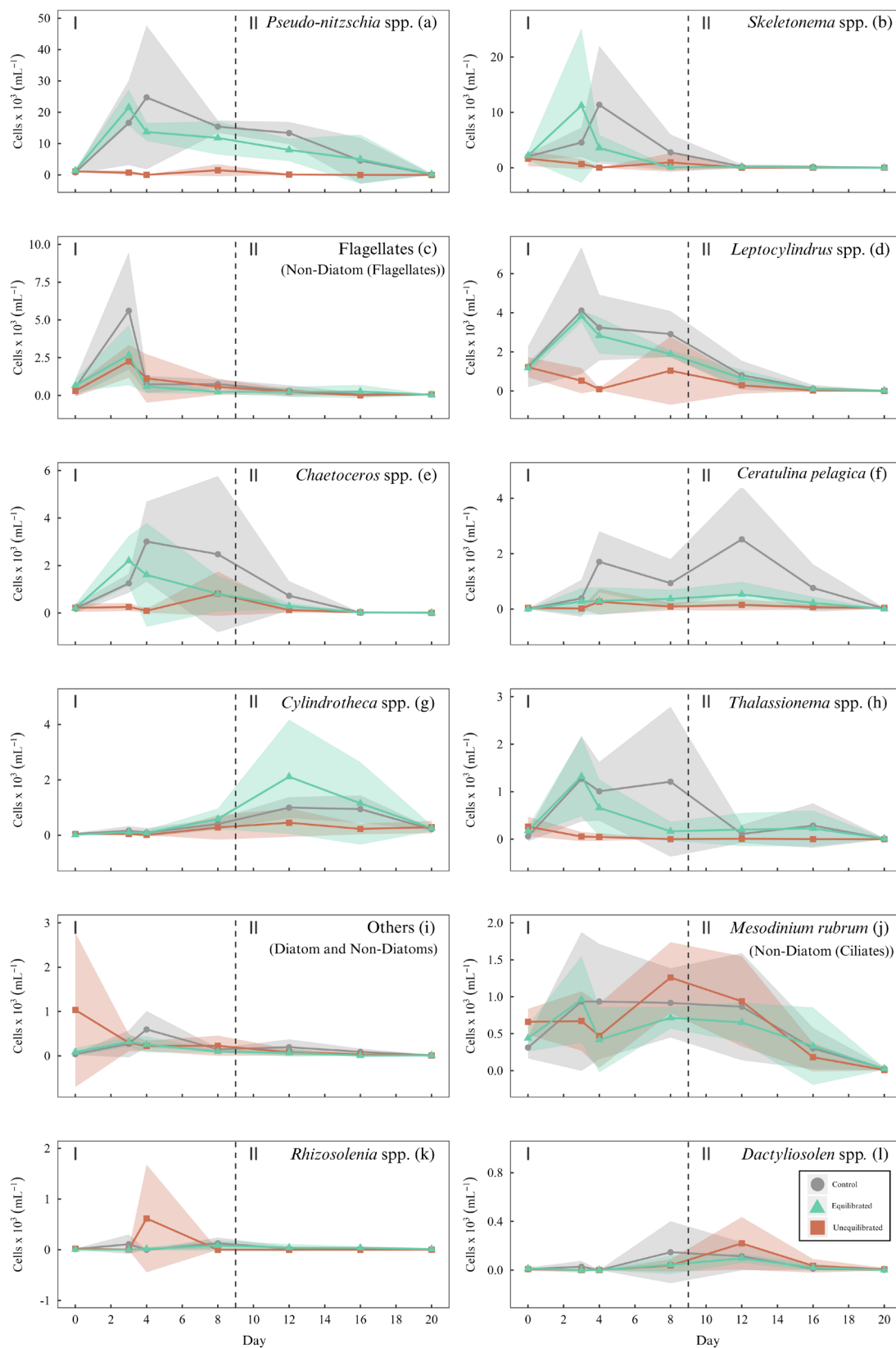


FIGURE 4 Abundances of phytoplankton groups obtained using light microscopy. (a) *Pseudo-nitzschia* spp.; (b) *Skeletonema* spp.; (c) Flagellates; (d) *Leptocylindrus* spp.; (e) *Chaetoceros* spp.; (f) *Ceratulina pelagica*; (g) *Cylindrotheca* spp.; (h) *Thalassionema* spp.; (i) Others; (j) *Mesodinium rubrum*; (k) *Rhizosolenia* spp.; (l) *Dactyliosolen* spp. Groups are listed in order of highest abundance (A. *Pseudo-nitzschia* spp.) to lowest abundance (L. *Dactyliosolen* spp.). Groups are diatoms unless otherwise stated. The dotted line denotes phases: I. pre-bloom and II. post-bloom. Standard deviation is represented by ribbon shading around the mean trendlines.

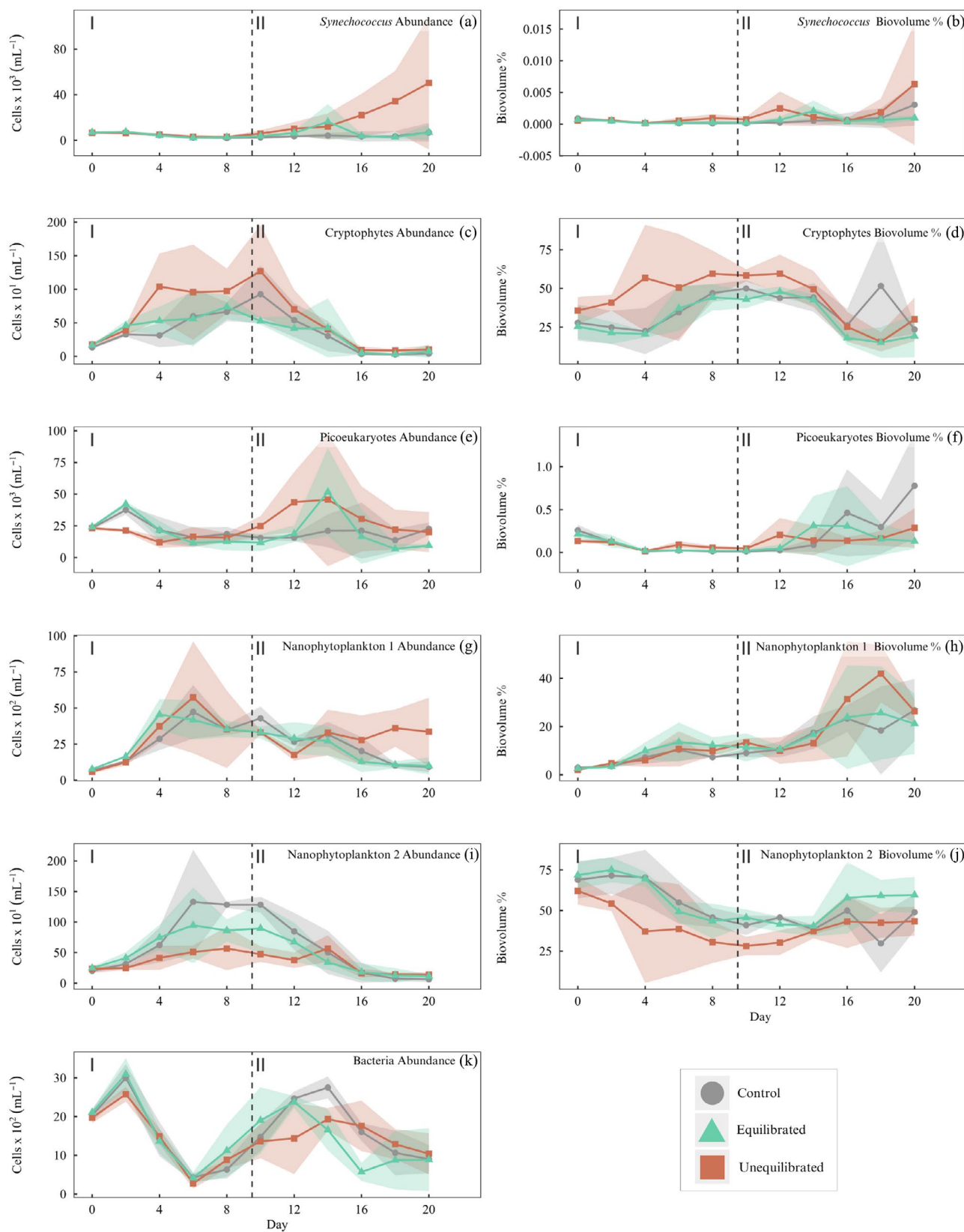


FIGURE 5 Flow cytometry phytoplankton abundance (left column) and biovolume percentage (right column). (a) *Synechococcus* abundance; (b) *Synechococcus* biovolume %; (c) Cryptophytes abundance; (d) Cryptophytes biovolume %; (e) Picoeukaryotes abundance; (f) Picoeukaryotes biovolume %; (g) Nanophytoplankton 1 abundance; (h) Nanophytoplankton 1 biovolume %; (i) Nanophytoplankton 2 abundance; (j) Nanophytoplankton 2 biovolume %; (k) Bacteria abundance. The dotted line denotes phases: I. pre-bloom and II. post-bloom. Standard deviation is represented by ribbon shading around the mean trendlines.

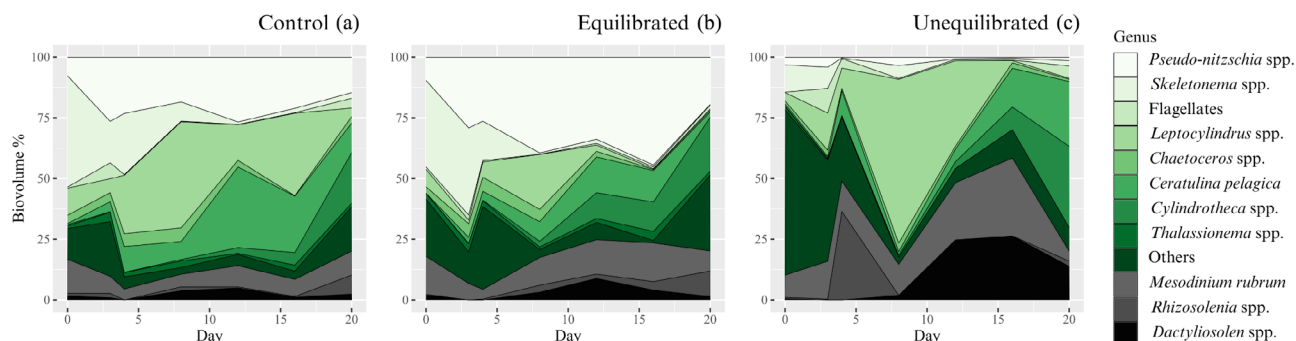


FIGURE 6 Phytoplankton biovolume percentage obtained via light microscopy. (a) Control; (b) Equilibrated treatment; (c) Unequilibrated treatment.

DISCUSSION

This study investigated the potential effects of hydroxide-based OAE on the composition of a coastal Tasmanian phytoplankton community. The discussion of the results focuses on the comparison between each treatment and the control, as it will provide information on the potential impacts of OAE in either its equilibrated or unequilibrated form. It is important to note that the microcosm design used in this experiment created a continuously high alkalinity perturbation. This allowed us to test a hypothetical OAE deployment where a community is exposed to an OAE-perturbed patch of seawater in a Lagrangian manner without dilution by unperturbed seawater. Here, the concentrations simulated in the microcosms would be representative of the center of a mixing zone, where the alkalinity perturbation is the highest and potential impacts are likely more pronounced. This scenario has limitations, as alkalinity addition in natural environments would dilute as it mixed into the surrounding area (Mu et al., 2023; Wang et al., 2023). As such, our experiment can be considered a rather extreme perturbation in which plankton were exposed to OAE-derived alkalinity (and associated carbonate chemistry changes) for theoretically longer than would happen in real-world applications. This should be kept in mind when evaluating the environmental effects observed here, as impacts would potentially be less pronounced in a real-world scenario in which dilution took place.

In both treatments, alkalinity increased by ~25% (Figure 3a). This increase in alkalinity would enable a 21% increase in the marine inorganic carbon sink (Schulz et al., 2023). Thus, it is crucial to evaluate the effects of OAE on plankton communities as observed here, against the 21% increase in the DIC sink and its considerable potential to counteract global warming (Ferderer et al., 2022). In other words, the very strong CDR potential established in the experiment was compared to the degree of plankton community change that was observed.

Delay of the phytoplankton bloom in the unequilibrated OAE treatment

One notable event during the experiment was the initial phytoplankton bloom, observed from the drawdown of DSi , NO_x^- , PO_4^{3-} , and the increase in Chl *a* between days ~3 and 8 (Figure 3f–i). Statistical analysis revealed a significant delay in the drawdown of DSi and PO_4^{3-} in the unequilibrated treatment, in which the OAE-induced pH_T increase (pCO_2 decline) was most pronounced (Figure 3b,c,f,h). The delay in nutrient drawdown was also observable in the delay and reduced buildup of Chl *a* in the unequilibrated treatment (Figure 3i). However, after the initially slower drawdown phase, the rate of nutrient decline and Chl *a* increase became similar in the unequilibrated treatment relative to the control and equilibrated treatment (nutrients; Figure 3f–h) or even a lag phase with no growth (Chl *a*; Figure 3i). Thus, the bloom delay in the unequilibrated treatment was not primarily due to generally lower rates of community nutrient uptake or growth but due to the induction of an initial slowdown and a delay of these processes. The slowdown phase was most likely induced by the pronounced and abrupt carbonate chemistry changes in the unequilibrated treatment, possibly due to a high pH_T effect or the associated rapid decline in pCO_2 . These abrupt changes may have had the following effect on the phytoplankton community: Either they slowed down community uptake and growth rates (physiological process) or they caused a reorganization of the phytoplankton community because some species were better equipped to cope with the changes than others (ecological process; note: a combination of both processes is also possible). A careful look at the data on community composition (Figures 4 and 5) suggested that the ecological process was more likely the predominant one. The major diatom species that provided much of the larger phytoplankton biovolume to the bloom (i.e., *Pseudo-nitzschia* spp., *Leptocylindrus* spp., and *Chaetoceros* spp., *Thalassionema* spp.; Figure 5) contributed to the bloom in the equilibrated treatment and control,

whereas they did not bloom in the unequilibrated treatment (Figure 4). Their collectively lower contribution to the bloom was also reflected in the flow cytometry data, in which the largest phytoplankton population identified with this method (nanophytoplankton 2) only contributed ~40% to the biovolume in contrast to ~70% in the equilibrated treatment and control (Figure 4). Instead of nanophytoplankton 2 (which was likely comprised the above-mentioned diatom populations), cryptophytes contributed most profoundly to the phytoplankton bloom in the unequilibrated treatment (Figure 5). Importantly, however, average Chl *a* concentration during the peak of the phytoplankton bloom was lower in the unequilibrated treatment than in the control and equilibrated treatment, suggesting that the shift to cryptophytes could not fully compensate for the failure of these diatoms to bloom. These results show that abrupt and relatively strong pH excursions (up to pH_T 8.7) induced through unequilibrated OAE (which exceeded the natural pH variation experienced by plankton in the Derwent Estuary, 7.83–8.36) can reorganize phytoplankton communities with potential knock-on effects for biogeochemical parameters (Figure 1a). However, the results also show that such reorganizations did not occur in the equilibrated treatment, which resembled the carbonate chemistry conditions that OAE needs to achieve on the order of years to be successful. As such, the reorganization of phytoplankton communities through unequilibrated OAE is a transient effect that only applies to locations where alkalinity increase is rapidly implemented.

Winners and losers in the plankton community under OAE

Effects on the plankton community were more pronounced in the unequilibrated treatment than in the equilibrated treatment. This is likely due to the stronger changes in the physiologically relevant carbonate chemistry parameters (pCO₂ and pH) that occurred in the unequilibrated treatment. When solely focusing on the analysis of the unequilibrated treatment (by comparing it to the control), we considered plankton groups as “winners” if they increased in abundance or biovolume or “losers” if they declined in abundance or biovolume.

Diatoms were abundant across all microcosms and throughout the entire experiment. However, distinct differences emerged in the responses of diatom genera between the control and two treatments. *Pseudonitzschia* spp., *Skeletonema* spp., *Leptocylindrus* spp., *Chaetoceros* spp., *Cerataulina pelagica*, and *Thalassionema* spp. were “losers” in the unequilibrated treatment (compared to the control) as they exhibited significantly lower abundances. Furthermore, these diatoms were the primary contributors to the

phytoplankton bloom, as indicated by the correlation between abundance, Chl *a* buildup, and nanophytoplankton 2 abundance data (the population most likely representing these genera in flow cytometry analysis; Figures 3 and 5). Due to the decline in abundances of these diatoms, other taxa counted microscopically, such as *Rhizosolenia* spp., *Mesodinium rubrum*, others, and *Dactyliosolen* spp., became relatively more important within the unequilibrated treatment. This was evident through the increased contribution of these genera to community biovolume (Figure 6).

Previous research on ocean acidification (OA) has identified winners and losers in phytoplankton communities under increased pCO₂ and decreased pH (Alvarez-Fernandez et al., 2018; Dutkiewicz et al., 2015; Schulz et al., 2017). Such conditions have generally been the opposite of what is observed under unequilibrated OAE (increased pH and decreased pCO₂); as such, it is expected that groups that benefit from OA may be negatively impacted by OAE. This is supported here with decreased abundances of *Skeletonema* spp. and *Chaetoceros* spp. in the unequilibrated treatment while previous work has revealed them to be winners under the high pCO₂ and low pH conditions accompanying OA (Bach & Taucher, 2019). In contrast, *Pseudonitzschia* spp. has been observed to display decreased abundances under elevated pCO₂ and pH as well as increased silicification to both elevated and decreased pCO₂, suggesting that this species may be relatively intolerant to any large change in carbonate chemistry irrespective of its direction (increase or decrease in pCO₂; Ferderer et al., 2024; Hoppe et al., 2017; Petrou et al., 2019; Tortell et al., 2008).

All diatoms defined above as “losers” showed an immediate decline in abundance following the addition of alkalinity in the unequilibrated treatment. Abundances remained significantly lower than those in the control and equilibrated treatments until the post-bloom phase, when they converged toward zero. Due to the immediate differences in abundances between the treatments, the negative response of these diatoms was likely a result of the decrease in CO₂ concentration and increase in pH associated with the unequilibrated OAE treatment. Both pH and CO₂ concentration are considered as the most important carbonate chemistry drivers controlling phytoplankton growth rates (Paul & Bach, 2020). Indeed, species-specific responses of diatoms to varying concentrations of pCO₂ and changes in pH have been well documented within the literature (Chen & Gao, 2003; Goldman et al., 2017; Riebesell et al., 1993). It has been thought that this variability in diatom responses is due to the diverse nature of this group with large differences in morphologies, size (2 μm – 2 mm), carbon fixation pathways, carbon concentrating mechanisms, and efficiencies for CO₂ (Hopkinson et al., 2011; Reinfelder et al., 2000). Carbon concentrating

mechanisms are often proposed as an explanation for species-specific differences in tolerances to changing carbonate chemistry conditions, as they enable the utilization of HCO_3^- under suboptimal conditions, generally when concentrations of CO_2 are below saturating concentrations for RuBisCO, a key enzyme for photosynthetic carbon fixation (Burkhardt et al., 2001; Rost et al., 2006). Furthermore, large differences in the efficiencies and pathways for carbon concentrating have been identified among species with some species using HCO_3^- as their primary carbon source while others rely on CO_2 (Rost et al., 2006; Trimborn et al., 2008, 2009). These species-specific characteristics likely contributed to the differing responses of diatom species observed here.

Picoeukaryotes showed decreased abundance in the unequilibrated treatment during the early stage of the experiment during which the bloom evolved (~45% decrease on day 2; Figure 5e). This was consistent with previous observations that reported Picoeukaryotes to be “losers” under unequilibrated OAE in a nearly identical setup conducted at a different time of year at the same location in Tasmania (Ferderer et al., 2022). The observation is also in accordance with a wealth of OA studies that have repeatedly observed picoeukaryotes as “winners” under high CO_2 and low pH, that is, the opposite conditions of the unequilibrated treatment (Bach et al., 2017; Davidson et al., 2016; Maugendre et al., 2017; Newbold et al., 2012; Sala et al., 2015; Schaum et al., 2012; Schulz et al., 2017; Thomson et al., 2016; White et al., 2020). As such, there is widespread evidence that OAE could reduce abundance of phytoplankton within the picoeukaryote population, thereby counteracting their stimulation through OA. The driver causing this response is putatively physiological, since experiments with *Micromonas* (a predominant picoeukaryote genus) have shown increasing growth and photosynthetic performance under OA (White et al., 2020).

Synechococcus, a Cyanobacterial genus in the same size class as picoeukaryotes (0.2–2 μm), showed a significant abundance increase in the unequilibrated treatment in the second half of the experiment (but please note the large variation between replicates). This observation is inconsistent with those by Ferderer et al. (2022), who observed a decline of *Synechococcus* under both equilibrated and unequilibrated OAE. Previous research on OA has not identified consistent responses of *Synechococcus* to carbonate chemistry changes (Sala et al., 2015; Schulz et al., 2017). It has been argued that observed treatment-specific differences in *Synechococcus* abundance were induced indirectly through differences in the grazing pressure (Bach et al., 2017). As such, changes in *Synechococcus* abundance merely reflected treatment effects on the grazer community. Indeed, a recent experiment testing environmental effects of mineral-based OAE observed

evidence of a reduction in grazer abundance, reflected in increased *Synechococcus* abundance (Guo et al., 2024). If the insights from this and past carbonate chemistry experiments hold true (i.e., *Synechococcus* responds variously, possibly depending on grazing), then it will be difficult to project *Synechococcus* responses to OAE, as responses will depend highly on the food web structure in any given system.

Ocean alkalinity enhancement did not significantly affect zooplankton abundance and biovolume sampled through net hauls, except for those of *Notiluca scintillans*, which showed relatively lower abundances in the unequilibrated OAE treatment. The limited effect of OAE via NaOH on the predominant copepod and cladoceran community was consistent with results from other mesocosm studies in Spain and Norway that reported no OAE effects on microzooplankton (Xin, Goldenberg, et al., 2024; Xin, Kittu, et al., 2024) copepods, jellyfish, and fish larvae (Goldenberg et al., 2024). Like our experiment, these studies used relatively high purity chemical products to simulate OAE, and changes in carbonate chemistry were the main driver for biological effect. (Note that Goldenberg et al., 2024 also established an OAE treatment with added silicic acid, and there was also no effect on zooplankton detected.) However, another microcosm study in Tasmania observed more widespread effects on zooplankton when OAE was implemented via the dissolution of dunite powder (Guo et al., 2024). In their study, Guo et al. (2024) argued that the effect on some zooplankton groups (mainly appendicularia) was driven by physical effects of lithogenic particles suspended in the water column, an effect that does not apply to the hydroxide-based OAE studied here. Collectively, the evidence available so far suggests that carbonate chemistry changes induced by OAE have limited effects on zooplankton, whereas other dissolution products that can arise when using less pure alkalinity sources (e.g., trace metals) have stronger effects.

The exception to this general observation was a negative response of the dinoflagellate *Notiluca scintillans* in the unequilibrated treatment (Figure 7a). *N. scintillans* is a 200–1000 μm phagotrophic harmful algal bloom species with global expansion (Harrison et al., 2011). In Tasmanian waters, *N. scintillans* was first reported in 1994 and today is a common zooplankton species observed in the Derwent estuary (McLeod et al., 2012) where we sourced our seawater for this experiment. The reason for the lower abundance in the unequilibrated treatment remains speculative. The pH_T was substantially higher in the unequilibrated treatment, but *N. scintillans* blooms occur in eutrophic water (Hallegraeff et al., 2019), associated with high productivity and, thus, naturally high pH levels. Previous studies have identified *N. scintillans* blooms in natural pH levels ranging from 7.49 to 8.69, yet these studies have not identified any significant correlation between

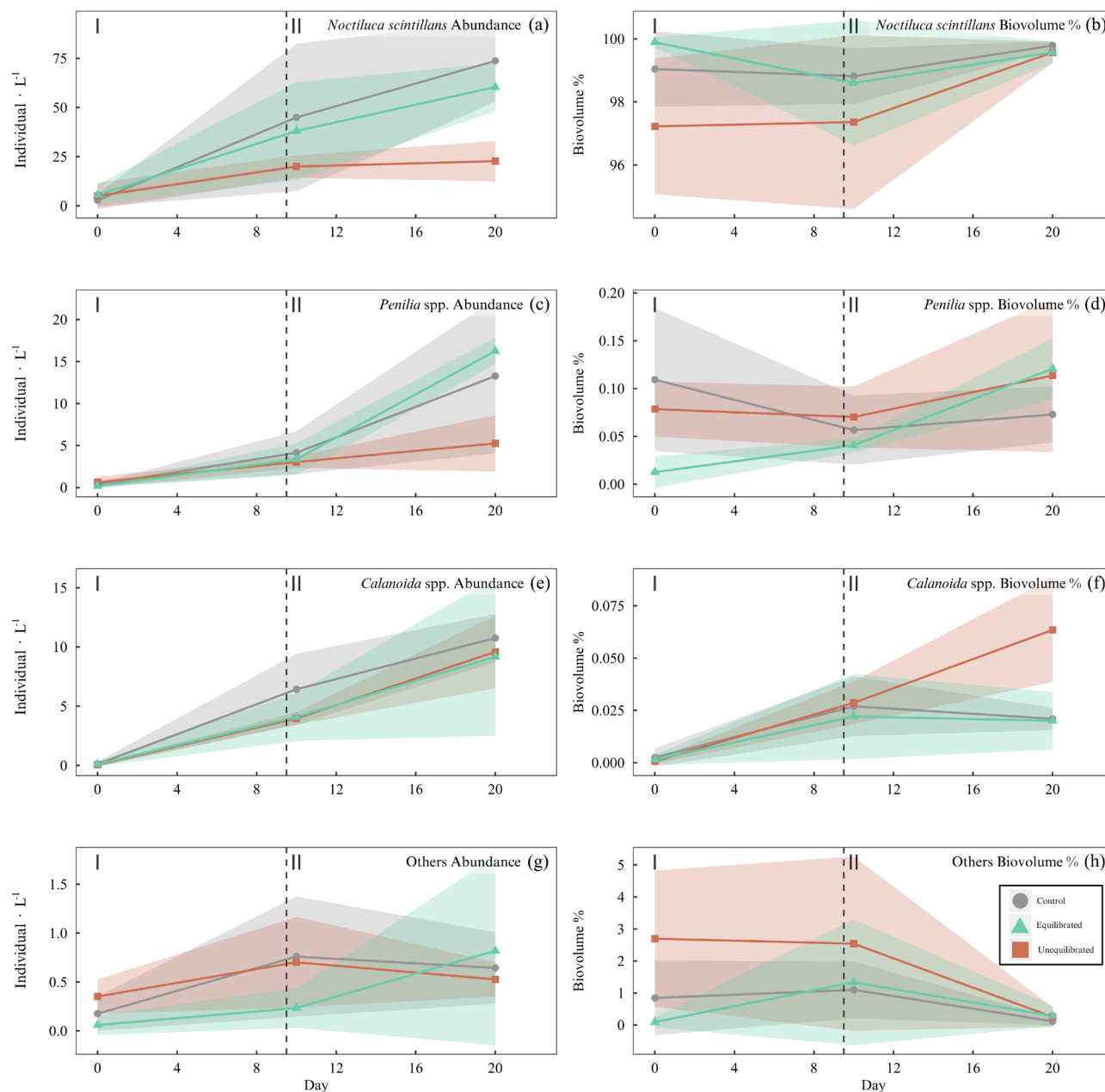


FIGURE 7 Light microscopy zooplankton abundance (left column) and biovolume percentage (right column). (a) *Noctiluca scintillans* abundance, (b) *Noctiluca scintillans* biovolume %, (c) *Penilia* spp. abundance, (d) *Penilia* spp. biovolume %, (e) *Calanoida* spp. abundance, (f) *Calanoida* spp. biovolume %, (g) Others abundance, (h) Others biovolume %. The dotted line denotes phases: I. pre-bloom and II. post-bloom. Standard deviation is represented by ribbon shading around the mean trendlines.

N. scintillans and pH (Rameshkumar et al., 2023; Turkoglu, 2013). A direct effect of pH (or other carbonate chemistry parameter changes associated with high pH, Figure 3) therefore seems less plausible. An indirect effect through reduced grazing success appears more plausible, since *N. scintillans* has a very specific prey preference (Hallegraeff et al., 2019), and the unequilibrated OAE treatment saw relatively pronounced changes in phytoplankton communities relative to the control and equilibrated treatment (Figures 4 and 5). In a previous study by Guo et al. (2024), *N. scintillans* benefitted from the OAE treatment that induced the

strongest changes of phytoplankton community composition, thereby further supporting this hypothesis. If true, the response of *N. scintillans* to OAE will be difficult to project, as it will depend on plankton community structure at the site of OAE perturbation. Nevertheless, the detrimental effect of hydroxide-based OAE on potentially harmful and invasive (in Tasmanian coastal waters) *N. scintillans* could arguably be seen as a co-benefit, rather than a risk, of unequilibrated OAE. Future studies should aim to identify the key drivers determining proliferation or suppression of harmful algal bloom species like *N. scintillans*, to generate predictive

understanding of how changes in the environment (including through OAE) would affect their occurrence.

CONCLUSIONS

Our study investigated the effects of hydroxide-based OAE on the composition of a coastal plankton community in Tasmania. Overall, we observed minimal effects of OAE when the increase of alkalinity coincided with atmospheric CO₂ equilibration, established by the addition of sodium bicarbonate, a scenario characterized by relatively limited changes in carbonate chemistry. The effects of OAE were more pronounced when the addition of alkalinity was not immediately equilibrated with atmospheric CO₂, a scenario characterized by relatively pronounced changes in pH and CO₂. Here, the predominant effects were: (1) a ~2-day delay in the onset of the main phytoplankton bloom (Chl *a* buildup) at the beginning of the study when inorganic nutrients were still abundant and (2) A reorganization in phytoplankton community composition observed in microscopic counts of some diatom species and flow cytometric determination of some phytoplankton groups. However, whether these moderate changes observed in the unequilibrated treatment in their entirety are beneficial or detrimental for marine food webs remains unclear. Overall, the degrees of change observed in the plankton communities appear limited when compared to the relatively extreme degree of OAE simulated here (~21% increase for seawater to store atmospheric CO₂). This is particularly true when considering that unequilibrated conditions that occur directly after hydroxide-based OAE treatment are a short-lived (weeks to years), transient state, as carbonate chemistry excursions toward low pH and CO₂ are rapidly mitigated through dilution and atmospheric CO₂ uptake (Mu et al., 2023; Wang et al., 2023; Zhou et al., 2024). Thus, largely consistent with previous studies (Ferderer et al., 2022, 2024; Goldenberg et al., 2024; Subhas et al., 2022; Xin, Kittu, et al., 2024), OAE implemented through hydroxides does, so far, not appear to have lasting negative effects on plankton communities. Upcoming globally replicated experiments following the same protocol as utilized here (Bach et al., 2024) will provide a way forward to consolidate or challenge this early impression.

AUTHOR CONTRIBUTIONS

Kiyas Kousoulas: Conceptualization (equal); data curation (lead); formal analysis (lead); investigation (lead); methodology (supporting); visualization (lead); writing – original draft (lead); writing – review and editing (lead).

Aaron Ferderer: Data curation (supporting); formal analysis (supporting); investigation (supporting); methodology (supporting); supervision (supporting); visualization (supporting); writing – original draft (supporting);

writing – review and editing (supporting). **Ruth Eriksen:** Formal analysis (supporting); methodology (lead); writing – review and editing (supporting). **Lennart T. Bach:** Conceptualization (equal); data curation (supporting); formal analysis (supporting); funding acquisition (lead); investigation (supporting); methodology (lead); supervision (lead); visualization (supporting); writing – original draft (supporting); writing – review and editing (supporting).

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

Figure S1. Flow cytometry gating methods. Plots a, b, and c come from microcosm M7 on day 4. (a) plot of fluorescence channels in sample; (b) gates for picoeukaryotes, nanophytoplankton 1, and nanophytoplankton 2; (c) gates for cryptophytes and *Synechococcus*; (d) Bacteria gate for microcosm M6 on day 20.

Figure S2. Flow cytometry gating results. Results come from microcosms M1 to M9 on day 2 (pre-bloom). The left column shows the controls (M1, M4, and M7), the middle column shows the unequilibrated treatments (M2, M5, and M8), and the right column shows the equilibrated treatments (M3, M6, and M9).

Table S1. List of phytoplankton in the sampled communities. Phytoplankton functional groups, genera, and/or species identified during light microscopy and scanning electron microscopy. Photos come from samples taken throughout the experiment.

Table S2. List of phytoplankton and zooplankton identified during light microscopy and scanning electron microscopy, including their volume formulas used for biovolume calculations.

Table S3. Average values and standard deviation of each treatment set including temperature, salinity, pH_T (pH total scale), TA (total alkalinity), pCO₂ (partial

pressure of carbon dioxide), DIC (dissolved inorganic carbon), Ω_{arg} (aragonite saturation state), DSi (dissolved silicate), NO_x⁻ (nitrate + nitrite), PO₄³⁻ (phosphate), and Chl a (chlorophyll a). A salinity measurement of 31.14 PSU was only taken once at the time of collection. TA data was measured every 4 days and remaining days were interpolated (interpolated days in red). Please note that TA measurements for microcosm 9 (equilibrated treatment) failed on day 0.5 (i.e., the sample taken directly after NaOH/NaHCO₃ addition) and microcosm 2 (unequilibrated treatment) on day 20. For those two datapoints, we used averages of the other two replicates from the same day for carbonate chemistry calculations. These measurements, and the interpolated data impacted by them, are denoted with a “*” in the table.

Table S4. Result of linear mixed effects models assessing the influence of treatments on zooplankton abundance.

Table S5. Post hoc test result assessing the variation in zooplankton abundance between genera within treatments.

Table S6. Post hoc test result assessing the variation in zooplankton abundance within a genus as a function of the treatment.

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